

Does Syntactic Structure Make Language Models More Robust to Noise?

LSTM vs. RNN Subject-Verb Agreement Under Training-Data Corruption

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Outline

- 1 Motivation
- 2 Background
- 3 Methods
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Motivation: robust grammar from noisy input

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- ① TODO: can injected SVA errors delay/prevent the hierarchical rule?
- ② TODO: is the linear rule equally learnable under noise?
- ③ TODO: do RNNG vs. LSTM differ in robustness?

Subject–verb agreement: hierarchical vs. linear

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Attractors as the diagnostic case

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Noise injection: five corrupted corpora

- Base: Linzen et al. (2016) SVA corpus — $\sim 1.58\text{M}$ Wikipedia sentences, vocab capped at the 10K most frequent types.
- Each sentence annotated with subject, verb, verb position, and $\#$ intervening attractor nouns.
- Build five training corpora, identical except agreement noise: flip the verb's number with probability p .

$$p \in \{0, 0.004, 0.02, 0.10, 0.50\} \quad (\text{baseline} \rightarrow \text{high})$$

- Flip via deterministic inflection (`lemminflect` + irregular copulas/auxiliaries); all other lexical/syntactic content preserved.
- Training-set size held constant $\Rightarrow$ isolates noise.

Leak-free evaluation protocol

- The five corpora are row-aligned with the source corpus.
- Reserve one fixed held-out set: 20k sentence indices (seed 1234).
- Those rows excluded from **every** training corpus $\Rightarrow$ no test leakage under any noise condition.
- Test items built from the **uncorrupted** source $\Rightarrow$ all test sentences are grammatical.
- Same test set for all conditions — only training noise varies.

- For each held-out sentence: a prefix, the grammatical verb v^+ , and its number-flipped counterpart v^- .
- Model is **correct** iff it prefers the grammatical form:

$$P(v^+ \mid \text{prefix}) > P(v^- \mid \text{prefix})$$

- Each pair tagged `has_attractor`: an intervening noun mismatching the subject in number.

- **Sequential baseline** — word-level LSTM LM (Linzen 2016 / Gulordava 2018 style): 2 layers, tied embeddings, dropout; trained from scratch per condition.
- **Structurally-biased** — RNNG (Dyer et al. 2016): explicit hierarchical structure.
- Core question: does a structural bias make hierarchical agreement *more robust* to noisy supervision?
- Fallback: ON-LSTM (Shen et al. 2019) — softer structural bias, no parse supervision needed.

- Primary: minimal-pair **accuracy** (grammatical $>$ ungrammatical).
- Secondary (graded): **surprisal** of the correct verb, $s = -\log_2 P(v^+ \mid \text{prefix})$.
- Split accuracy by attractor presence; reliance on the linear cue = **attractor gap**

$$\Delta = \text{acc}_{\text{no-attr}} - \text{acc}_{\text{attr}}$$

- Small $\Delta \rightarrow$ hierarchical rule; large $\Delta \rightarrow$ nearest-noun (linear) rule.
- Averaged over seeds $\{1, 2, 3\}$; central analysis: Δ vs. noise.

Accuracy degrades with noise

TODO: `acc_vs_noise.png`

The attractor gap widens with noise

TODO: attractor_gap.png

TODO: `acc_by_attractor.png`

Learning trajectories: delay vs. prevention

TODO: acc_trajectory.png

Does structure buy robustness?

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Takeaways and future work

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Thank you!

Questions?